

Positive Evidence Is Not Target Evidence

Invariant Evidence-Ratio Attention for Multi-Label Few-Shot Learning

Short research handbook and pilot protocol

Status. This is a testable proposal, not a claim of completed results. It keeps the idea that a support image is a mixture, but rejects literal feature subtraction. The central question is: *which visual evidence is enriched in target-positive supports relative to target-negative supports, and remains enriched when the support mixture changes?*

1 Problem and motivation

For a target pathology c , a positive chest radiograph may also contain devices, co-occurring diseases, acquisition marks, or site-specific texture. A positive-only prototype,

$$p_c^+ = \frac{1}{|S_c^+|} \sum_{x \in S_c^+} f_\phi(x),$$

therefore represents “everything common in c^+ supports”, not necessarily evidence of c . With only a few supports, a chest tube can become the prototype for pneumothorax, or cardiomegaly can become the prototype for edema.

The first idea was to remove named co-labels,

$$R_c(x) = T(x) - \text{Explain}(T(x), Y_x \setminus \{c\}).$$

That formulation should be rejected:

Failure	Consequence
Incomplete co-labels	Unannotated findings and artifacts remain in the residual.
Entangled features	Subtracting a device may also remove pathology evidence.
Near-perfect co-occurrence	Target and nuisance are observationally non-identifiable.
Per-label explanation	Cost grows with the number of observed co-labels.
Target heterogeneity	Over-aggressive invariance may suppress genuine subtypes.
Noisy target status	A false negative contaminates the reference evidence bank.

2 Proposed method: IERA

Minimal premise. Use only reliable target status c^+ or c^- ; every other label may be missing and is treated as unknown. Let a frozen ViT produce patch tokens z , followed by a small learned projection h_θ . Divide supports into two or more mixture environments e , such as different co-label panels, datasets, sites, or views. Each environment provides small positive and negative token banks $Z_{c,e}^+$ and $Z_{c,e}^-$.

Step 1: positive-versus-negative evidence ratio.

$$\text{LME}(z, A) = \log \left[\frac{1}{|A|} \sum_{u \in A} \exp \left(\frac{\cos(h_\theta(z), h_\theta(u))}{\tau} \right) \right], \quad r_{c,e}(z) = \text{LME}(z, Z_{c,e}^+) - \text{LME}(z, Z_{c,e}^-).$$

The ratio is high only when a patch resembles positive evidence more than negative evidence. Positive self-matches are removed whenever possible.

Step 2: retain evidence that survives a mixture change.

$$R_c(z) = -\beta \log \left[\frac{1}{E} \sum_{e=1}^E \exp \left(-\frac{r_{c,e}(z)}{\beta} \right) \right] = \text{softmin}_{e \in \{1, \dots, E\}} r_{c,e}(z) \leq \min_e r_e(z) + \beta \log E,$$

the two are separated when $\delta > \beta \log E$. This is a conditional robustness statement, not a causal-disentanglement theorem.

Hard boundary. If c and nuisance d always co-occur, have inseparable visual support, and never vary across observed environments, no observational method can identify which one is the target. IERA also cannot fix missing or incorrect target labels, and its attention map is predictive evidence rather than guaranteed anatomical localization. The robust claim needs $E \geq 2$; $E = 1$ reduces to ordinary contrastive ratio attention.

3 Training and inference

Meta-training episode.

1. Hide one base pathology c and permute its episode identity.
2. Sample K^+ verified-positive and K^- verified-negative supports in each of E mixture environments. Unknown and uncertain target labels are unused.
3. Cache frozen-ViT patch tokens, construct p_c , and score positive and negative queries.
4. Optimize query binary cross-entropy,

$$\mathcal{L}_{\text{episode}} = -\frac{1}{|Q|} \sum_{q \in Q} [y_q \log \hat{y}_c(q) + (1 - y_q) \log(1 - \hat{y}_c(q))].$$

Initially train only $h_\theta, \tau, \tau_a, \tau_q, \beta$, and γ .

Meta-test. For a held-out pathology, construct its prototype from few verified c^+ and c^- supports without a gradient update. No co-label list or per-co-label explanation module is required. One cached ViT pass plus batched token similarities is sufficient.

4 Pilot experiment

Data. Use patient-disjoint splits. Start on MIMIC-CXR and CheXpert; reserve NIH ChestX-ray14 for external validation after harmonizing label definitions. Treat blank or uncertain labels as unknown, never as negative. Candidate stress tests are pneumothorax-support devices, edema-cardiomegaly, effusion-atelectasis, and pneumonia-consolidation. Use a pair only if all four $(c, d) \in \{0, 1\}^2$ strata have at least 50–100 patients.

Paired support-composition test. Hold q, c, K fixed and replace a d^+ -dominated positive panel with a d^- -dominated panel:

$$\text{SMS}_{c,d} = \mathbb{E}_q |\hat{y}_c(q; S_{c,d^+}) - \hat{y}_c(q; S_{c,d^-})|.$$

This isolates sensitivity induced by the support panel. Match panels on dataset, view, and available demographics; report residual imbalance.

Baselines.

- Multi-label ProtoNet and a frozen-ViT linear probe.
- Patch attention / nearest-neighbor prototype and ProPOT-style local selection.
- Matched-support ProtoNet, the most important sampling-control baseline.
- Label-conditioned projection, adversarial nuisance invariance, and GroupDRO.
- IERA with $E = 1$, without negatives, without the soft minimum, and with an unfrozen encoder.

Report. AUROC and AUPRC; worst-stratum performance; SMS; false-positive activation on c^-d^+ ; Brier score / ECE; runtime; and patient-bootstrap confidence intervals. Evaluate $K \in \{1, 3, 5, 10\}$ while acknowledging that the full invariant mechanism is not identifiable at one shot.

Go/no-go gate. Continue only if a standard prototype shows a reproducible support-swap shift on at least three target-confounder pairs and a second dataset; IERA should reduce SMS by at least 30%, improve worst-stratum AUROC or AUPRC by at least 2 points, and lose less than 1 point on ordinary random-support evaluation.

5 Contribution and novelty boundary

The intended contributions are:

1. a fixed-query, paired support-composition stress test for incomplete-label medical FSL;
2. evidence-ratio attention that asks what distinguishes c^+ from c^- , rather than what merely repeats among positives;
3. environment-robust prototype induction with an explicit identifiability boundary and cross-dataset CXR validation.

Do *not* claim the first method for mixed multi-label supports, feature disentanglement, or spurious FSL. LaSO already studies feature-space label-set difference; ProPOT selects target-relevant regions from mixed supports; FewSTAB and MetaCoCo study spurious few-shot behavior. The defensible novelty is their intersection: *novel-label prototype induction under controlled support co-label shift using a cross-environment positive/negative patch evidence ratio*. A full Scholar/Scopus novelty audit is still required before submission.

References

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